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E-230 CABLES AND WIRES

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, instruction, commissioning and testing of all cables and wires for power, control and instrumentation cables in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 National

- | | | |
|----------|---|--|
| PUIL | : | Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation –Indonesia). |
| SPLN | : | Standard Perusahaan Umum Listrik Negara. |
| SII 0618 | : | Standard Industri Indonesia Kabel Telepon Tanah Tanpa Perisai Berisolasi dan Berselubung Polietilen Berisi Petrojelli. |
| SII 0710 | : | Standard Industri Indonesia Kabel Telepon Rumah Berpelindung Elektris Berisolasi dan Berselubung P.V.C. |

2.2 Institute of Electrical and Electronics Engineers

- | | | |
|----------------|---|---|
| ANSI / IEEE 48 | : | Test Procedures and Requirements for High-Voltage Alternating-Current Cable Terminations. |
| ANSI / IEEE 82 | : | Test Procedure for Impulse Voltage Tests on Insulated Conductors. |

2.3 National Electrical Manufacturers Association

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|------------|---|-----------------------------------|
| NEMA WC 30 | : | Color Coding of Wires and Cables. |
|------------|---|-----------------------------------|

2.4 International Electrotechnical Commission

- | | | |
|-----------|---|---|
| IEC 60085 | : | Electrical insulation – Thermal classification. |
| IEC 60228 | : | Conductors of insulated cables. |
| IEC 60230 | : | Impulse tests on cables and their accessories. |

IEC 60245	:	Rubber insulated cables-Rated voltage up to and including 450/750V.
IEC 60287	:	Electric cables-Calculation of the current rating.
IEC 60332	:	Tests on electric cables under fire conditions.
IEC 60502	:	Power cables with extruded insulation and their accessories for rated voltage from 1 kV (Um=1.2 kV) up to 30 kV (Um=36 kV).
IEC 60584-3	:	Thermocouples-Extension and compensating cable Tolerances and identification system.
IEC 60811	:	Common test methods for insulating and sheathing materials of electric cables and optical cables.
IEC 60853	:	Calculation of the cycle and emergency current rating of cables.
IEC 60885	:	Electrical test methods for electric cables.
IEC 60986	:	Short-circuit temperature limits of electrical cables with rated voltages from 6 kV (Um=7.2 kV) up to 30 kV (Um=36 kV).

3. OPERATING AND DESIGN CONDITIONS

Power, control, instrumentation, communication, thermocouple extension and special application cables and wires shall have a minimum design life of 30 years when used under the following operating and design conditions.

(1) Operating Voltages

- a. Medium voltage power cable: 6.3 kV, 3 wire, 50 Hz. The system neutrals shall be low-resistance grounded.
- b. Low voltage power control and instrumentation cables : Shall be 400V, 3 or 4 wire, 220 V, 2 wire, 50 Hz with the system neutrals solidly grounded and ungrounded 220 V DC.
- c. Thermocouple extension cables: normal operating electromotive force (EMF) shall be in accordance with ASTM E230 or equivalent.
- d. System operating voltage level shall be in accordance with Technical Specification E-001.

(2) Operating Temperatures

The maximum continuous ambient temperature will be 50°C.

The maximum continuous operating temperature of conductor shall be 90°C. An emergency overload temperature of 130°C and a maximum short circuit temperature of 250°C will be permitted for durations and frequencies as specified in the referenced IEC standards.

(3) Load Factor

The maximum load factor shall be 100%.

(4) Installation

The cables shall be suitable for indoor and outdoor installations exposed in air, in conduits, in cable trays, in underground concrete encased steel or plastic conduits, in underground trenches and for direct burial. Unarmored cables shall be suitable for installation in steel conduits.

(5) Environmental Conditions

The meteorological, seismic and environmental conditions for the Site are as indicated in Part 4, Subsections 1.1, 2.14 and 2.16.

4. DETAILED REQUIREMENTS

4.1 General

Cables and wires shall be manufactured or assembled in Indonesia.

All terminals and wires shall have identifications as shown on the Contractor's schematic and wiring diagrams. Wiring between equipment, or equipment and terminal blocks mounted on the same skid or shipping section shall be continuous with no splices.

Single conductor control cables shall not be run in cable trays. Cables outside of an enclosure or junction box shall be protected by conduit.

4.2 Cable Construction

The cables shall be single conductor or multi conductor assemblies, as specified.

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4.2.1 Conductors (core)

All conductors, except thermocouple extension cables, shall be Class 2, annealed stranded copper in accordance with IEC 60228.

Thermocouple extension cable conductors shall be solid and conductor materials shall be as per ASTM E 230 or equivalent.

The medium voltage power cable conductors shall be provided with a layer of extruded semi-conducting compound to meet with IEC 60502.

4.2.2 Insulation

The insulation of low voltage power, instrument, control and communication cables shall be polyvinyl chloride (PVC) and/or cross-linked polyethylene (XLPE), fire retardant, with low acid gas evolution.

The insulation of medium voltage cables shall be a high quality non-flame propagating ozone resistant extruded solid dielectric of cross-linked polyethylene (XLPE).

For high temperature (125-150°C) applications silicone rubber or Tefzel insulation may be used subject to approval by the Owner.

4.2.3 Conductor Jacket (Inner Sheath)

All single and multi conductor power cables shall be provided with a fire retardant and low acid and gas evolution jacket over the complete cable assembly.

4.2.4 Bedding (inner covering and fillers)

Multi conductor cables shall have bedding, which shall be suitable for the specified operating temperature and insulating material.

4.2.5 Armor

All multi-conductor cables shall be provided with continuous, impervious, close fitting, galvanized steel corrugated tube or tape or wire armor unless otherwise indicated or accepted in writing by the Owner. The armor grounding shall be in accordance with Technical Specification E-241,

4.2.6 Outer Jacket

All cables shall be provided with a fire retardant and low acid gas evolution outer jacket over the inner sheath, screen or the armor. This jacket may be polyvinyl chloride (PVC) provided it is certified by recognized independent international authority.

All cables shall be permanently identified by permanent surface printing or indenting (not to exceed 15% of the outer jacket thickness) to show the following:

- (1) Manufacturer's name
- (2) Rated circuit voltage
- (3) Conductor size
- (4) Number of conductors
- (5) Insulation material (not trade name)
- (6) Jacket material (not trade name)
- (7) Year of manufacture

The distance between the end of one (1) complete set of marks and the beginning of the next shall not exceed 500 mm.

4.2.7 Cable Splice and Rework

Factory cable splices and/or rework of the insulation and jacket is prohibited unless agreed to by the Owner in writing, in which case the following shall apply:

- (1) Splices and/or rework of the insulation and jacket may be required during manufacturing to eliminate defects, obtain specified shipping lengths, etc. Any such rework of splices shall conform to the material, dimensional and test requirements of the cable. Conductors and cables which have been reworked or spliced shall be tested as required for the specified cable and manufacturer shall include with his test reports the results of any tests for such rework and splices.
- (2) Materials for splices and rework shall be controlled and documented in accordance with the quality assurance/quality control requirements.

Field cable splicing and/or rework is prohibited.

4.3 Specific Requirements

4.3.1 Power Cables

- (1) Medium voltage power cables shall be single or three-conductor cable with the following voltage ratings:
6 kV System: 3.6/6 (7.2) kV
- (2) Low voltage power cables shall be single, two, three or four conductor cable with a voltage rating of 0.6/1 (1.2) kV.
- (3) The minimum conductor size for power cables shall be 4.0 mm².
- (4) All power cable conductors shall be identified by coloring the individual conductor jackets as follows:

<u>Number of</u> <u>Conductor</u>	<u>Phase</u>	<u>Letter</u> <u>Designation</u>	<u>Color</u>
Three	1,2,3	R, S, T	Red, Yellow, Black
-	Neutral	N	Light Blue
-	Insulated Ground	G	Green-Yellow

- (5) The outer jacket shall be colored red for medium voltage cables and black for low voltage cables.

4.3.2 Control Cables

- (1) Control cables shall be rated 600 volts.
- (2) The minimum conductor size of control cables used for field connections shall be 2.5 mm². For electronic voltage levels and connecting to electronic devices within cabinets, the minimum control cables shall be 0.5 mm².
- (3) The minimum conductor size for current transformer secondary circuits shall be 6.0 mm². The actual cable size shall be determined by the VA or voltage drop requirements of the application. The cable sizing calculation shall be submitted by the Contractor for the Owner's approval.
- (4) The maximum number of conductors in a multi-conductor control cable shall not exceed thirty (30) without the Owner's approval.
- (5) The insulation of all individual conductors shall be the same color (black, white or natural), with surface printing of numbers in sequence, in accordance with relevant standards.
- (6) The outer jacket shall be colored black.

4.3.3 Instrumentation Cables

- (1) Instrumentation cables shall be rated 600 volts, and shall be shielded twisted pairs or twisted triads.
- (2) The minimum conductor size of instrumentation cables shall be 1.5 mm².
- (3) Each twisted pair or triad shall be provided with a copper or aluminum Mylar shield and a tinned stranded bare copper drain wire. The individual pair shields and drain wires shall be insulated from adjacent pairs and the overall shield.
- (4) Each pair or triad shall be twisted with a lay of approximately 50 mm.
- (5) Each multiple pair or triad cable, in addition to an individual shield for each pair or triad, shall be provided with an overall shield of copper or aluminum Mylar and one drain wire.
- (6) Pair, triad and overall shields shall have a layer of aluminum or copper 0.025 mm thick and a layer of Mylar 0.05 mm thick.
- (7) The minimum drain wire conductor size shall be 1.0 mm² for all instrument cables. The drain wire shall be in intimate contact with the metal part of its particular shield.
- (8) The maximum number of pairs or triads in a multiple pair or triad shall be twelve (12) pairs or triads.
- (9) For pairs, one (1) conductor shall be black and one (1) white. For triads, one (1) conductor shall be black, one (1) white and one (1) red. The pairs of multi-pair or the triads of multi-triad cables shall be identified with surface printing of numbers in sequence, on each conductor of each pair or triad (same number of each conductor of a given pair or triad), in accordance with relevant standards.
- (10) The outer jacket shall be colored white.

4.3.4 Thermocouple Extension Cables

- (1) The cables shall be rated for 600 V and shall consist of single or multiple-twisted pairs with individual drain wires and shields (aluminum or copper Mylar). The cables shall also have an overall shield, drain wire and jacket and be of similar construction to instrument cables, Clause 4.3.3 except multiple-pair thermocouple extension cables may use a minimum conductor size of 0.75 mm².

- (2) The thermocouple extension cable conductor materials shall be thermocouple materials as required for the application in accordance with ASTM E 230; e.g. chromel-constantan, chromel-alumel, etc.
- (3) The colors of insulation shall be in accordance with ASTM E 230 and as follows:

Color of Insulation

<u>Cable Type</u>	<u>Positive - Negative</u>	<u>Color of Overall Jacket</u>
Chromel-Alumel	Yellow - Red	Yellow
Chromel-Constantan	Purple - Red	Purple

- (4) High temperature cables shall have insulation suitable for the maximum temperature to which it may be subjected in service. Temperature classification of insulation shall be in accordance with IEC 60085.
- (5) Multi-pair cables shall consist of pairs of conductors with colors as indicated above. The pairs shall be identified by surface printing of numbers in sequence on each conductor of each pair (same number on each conductor of a given pair), in accordance with relevant standards.

4.3.5 Special Application Cables

- (1) The Contractor shall provide all special application cables required for various equipment provided as part of the Work.
- (2) Cables for communication system shall be in accordance with relevant IEC.

4.4 Cable Sizing Criteria

The Contractor shall use the following criteria for selecting and sizing the cables.

4.4.1 Voltage Drop in Feeders

The total voltage drop in AC or DC feeder and branch circuits shall not exceed 3% of bus nominal voltage.

The voltage drop calculation for equipment shall be based on the rated nominal voltage of the distribution bus feeding the equipment, full load current of the equipment being fed at its rated voltage and maximum feeder cable impedance.

The voltage drop in control circuit wiring shall not exceed a value that may cause contactors to drop out, or cause auxiliary relays or contactors to fail to pick up or

the control system to malfunction in other way under any auxiliary system voltage condition, including during the starting of large motors.

4.4.2 Cable Ampacity (continuous current carrying capacity)

The cable ampacity shall be derated based on the method of installation and groupings in accordance with the cited standards as follows:

- (1) Cables in concrete duct banks shall be calculated in accordance with IEC 60287.
- (2) Cables in cable trays shall be sized in accordance with IEC standards.
- (3) Cables in conduits shall be sized in accordance with IEC 60287.
- (4) Other derating factors which shall be considered are as follows:
 - a. The ambient temperature at the termination point shall be considered to be 50°C.
 - b. 10% derating factor due to current unbalance when two (2) or more conductors per phase are used.
 - c. The Contractor's cable ampacity calculations shall be submitted to the Owner for review.
- (5) All cables shall have adequate conductor sizes to carry the full starting current of the motors, during the period of motor acceleration without exceeding the specified short duration conductor temperature of 130°C.
- (6) For motor circuits, the cable shall have a current capacity of at least equal to 115 % of full load current rating of the motor after application of the appropriate cable derating factors.

4.4.3 Short Circuit Current Capacity

The cable short circuit rating shall be based on the available fault currents at the associated voltage level as given in Technical Specification E-001.

The cable short circuit withstand time shall not be less than the total operating time, including the operating time for the protective relays (including back up relay time) and the circuit breaker fault clearing time with a minimum of one (1) second withstand rating.

Under worst conditions, the conductor temperature shall not exceed its maximum short circuit temperature of 250°C.

The Contractor's calculations for cable short circuit current capacity shall be submitted to the Owner for review.

4.5 Cable Installation

4.5.1 General

All cables routed through the cable trays shall be installed as follows:

- (1) All cables shall be clamped as close as possible to the points of entry to or exit from cable trays.
- (2) Single conductor power cables shall be laid in cable trays in parallel runs in trefoil configuration and tied together with cable tie wraps to maintain the formation.
- (3) Cables installed in vertical or inclined cable trays runs shall be fastened to cable tray rungs with suitable cable ties not more than one (1) meter a part.
- (4) Cables installed in vertical conduit risers shall be supported with cable grips not more than ten (10) meters a part.
- (5) All cables entering or leaving enclosures shall be provided with watertight insulated bushings.

4.5.2 Preparation of Cable Trays and Trenches before Installing Cables

The route of each cable pull shall be inspected to ascertain locations where the cables may be subject to abrasion or unusual strain. Workmen shall be stationed at these locations during the pulling operation to guard against damage to cables.

Cables shall not be pulled around corners or obstructions without the use of cable guides of the proper radius.

4.5.3 Preparation of Conduits and Ducts before Installing Cables

The inside edges of conduits or ducts shall be reamed smoothly to prevent any damage to the cable sheath or jacket.

4.5.4 Cable Pulling Equipment

A suitable flexible feeder tube, cable protectors or sheaves shall be used to protect and guide the cable from the cable reel through manholes and into conduit or duct.

Pulling rope used may be either nylon, polypropylene or manila rope. Bare wire rope shall not be used.

A dynamometer/tensionmeter shall be used for monitoring cable pulling tension in all cases where a manual or power-operated winch or cable-pulling machine is used. A dynamometer/tensionmeter is not required for hand pulls.

4.5.5 Cable Pulling and Installation

When pulling cables through conduits, cable trays and ducts that are supported by rods or angle hangers, they shall be braced to withstand the tensions used for pulling the cable.

Only cable pulling lubricants compatible with cable jacket should be used when pulling cable through conduits. The choice of lubricant shall be approved by the cable manufacturer.

When pulling cable into conduit, ductruns, etc., the cable shall be fed straight into them by hand. For large conductors, sheaves shall be used to avoid short bends, sharp edges and “crossovers.” The sheave radius shall be at least the minimum pulling radius of the cable.

When cables are routed from a manhole or a handhole through conduit nipples, the cables shall be inspected both inside and outside the manhole or handhole to ensure that slack cable is provided to prevent cable damage from strain at the conduit entrances or exits.

All cables routed through cable support sections in cable tray risers shall be properly supported with cable grips.

In the cable manufacturer’s minimum cable bending recommendations are not available, the following minimum cable bending data excerpted from NEMA/ICEA S 68-516, appendix L.

4.5.6 Pulling Tensions

The maximum permissible pulling tensions when using either a pulling eye attached to all conductors or a basket weave grip over all conductors shall not be exceeded.

For medium and high voltage cable, the pulling tension shall be directly upon the conductor and not upon the jacket or insulation.

For low voltage power, control or instrument cable, the pulling tension shall be uniformly distributed between the conductors. The pulling tension may be exerted either on the bare conductor or the jacket and insulation.

4.5.7 Protecting and Supporting Cables

Cable ends shall be sealed prior to pulling through manholes, ductruns, conduits or areas where cables may be subjected to moisture during pulling. Cable ends shall not be left unsealed for any cable on reels.

Cables that are pulled through a manhole shall be trained along the sides of the manholes and temporarily supported if necessary. Such cables shall be provided with sufficient slack for proper arrangement on racks, hangers or trays where future splices can be made.

4.5.8 Cable Splices

No cable splices shall be permitted unless approved in writing by the Owner any splices shall be made only at locations and in the cables specified in the Contractor's Drawings.

In general, splices in cables which are installed in concrete duct banks or trenches in earth shall be made only in manholes or handholes provided for splices.

4.5.9 Accounting for Cable

The Contractor shall keep a running account of the use of power, control an instrument cable in accordance with the Specification.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Cable Tests and Test Reports

Tests shall be performed by the Contractor and certified test reports shall be submitted to the Owner to confirm compliance with the Specification and Standards.

Tests shall consist of manufacturing tests (routine and type tests) and tests after electrical installation, as specified.

5.2 Manufacturing Tests

5.2.1 Routine Tests

Routine tests shall be performed on all cables, in accordance with IEC 60228, IEC 60502 and IEC 60811, or equivalent.

5.2.2 Type Tests

Type tests shall be performed on production samples for each type and voltage rating in accordance with IEC 60228, IEC 60502 and IEC 60811, or equivalent.

The Contractor may submit certificate of type tests that have been performed on identical cables with identical voltage ratings. Notwithstanding any provisions in a standard, the Owner reserves the right to accept or reject such certificate in lieu of the specified tests.

The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or are rejected.

5.2.3 Flame Propagation Type Tests

Vertical tray gas-burner flame propagation tests shall be performed in accordance with IEC 60332, or equivalent.

5.3 Test after Installation

The Contractor shall perform continuity and insulation resistance tests on all cables after installation.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the cables and wires shall conform to the requirements of Part 4, Subsection 2.7. The procedure specified shall include, but not be limited to, the followings:

- (1) Factory splice and rework.
- (2) Flame propagation.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Routine Test Reports
- (2) Flame propagation test reports

7. CABLE INFORMATION SYSTEM

7.1 Cable Pulling Progress

The Contractor shall keep the Owner fully informed of cable pulling progress by means of an appropriate Cable Information System (CIS) which shall include at least the following information:

- (1) Cable number
- (2) Cable description
- (3) Service: Power feed or control or function of the cable
- (4) Grounding Cable/Bar Size: For motors and other equipment that require definite grounding cable/bar.
- (5) Detailed Block Interconnection Drawing Number: Where the equipment and interconnecting cable is shown.
- (6) “From” Equipment Description.
- (7) “To” Equipment Description: This shall indicate the description of equipment where the cable is going to. This information shall be the same as the “From” Equipment Description.
- (8) Routing : This shall indicate the routing of cable through various section of cable trays (trenches), cable risers, conduits, underground concrete duct runs and manholes. The sections of cable trays (trenches), risers, duct runs and manholes shall be identified by unique identifying numbers, which have been assigned by the Contractor and shall be shown on physical drawings.
- (9) Estimated Length: This shall indicate the estimated length of cable based on the routing.
- (10) Actual Length : Shall indicate the actual length of cable pulled.
- (11) “From” Cable Termination : Shall indicate whether the cable is terminated at the “From” end equipment.
- (12) “To” Cable Termination: Shall indicate whether the cable is terminated at the “To” end equipment.
- (13) Each cable tabulation page shall have a revision column

7.2 Final Cable Report

Prior to commissioning of each Gas Turbine or Steam Turbine and a combined cycle power plant, a final report shall be issued covering the above data for the total associated cable system.

7.3 Report Format

Prior to routing or pulling any cable, the Contractor shall provide complete format and details of the cable information system for the Owner's review.

(Last page of E-230)